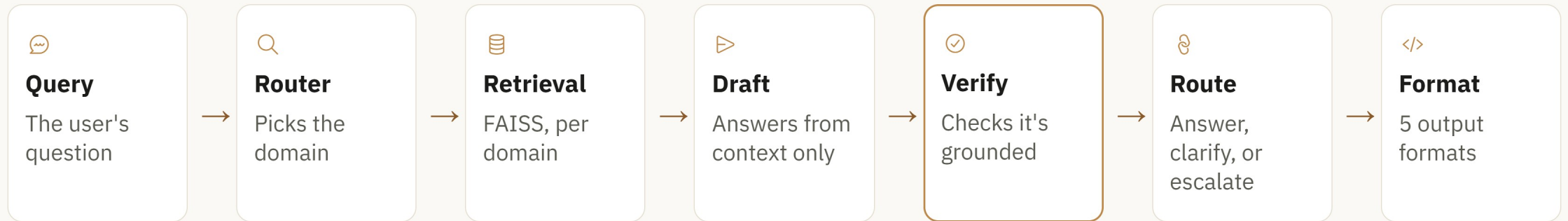


How a query becomes a trusted answer



Multi-turn memory carries the last few exchanges with every query, so a follow-up like "what about parental leave?" resolves against the previous question instead of starting over.

Built to run on nothing but the free tier

LLM	Gemini gemini-3.5-flash-lite, via google-genai
Embeddings	sentence-transformers, all-MiniLM-L6-v2 — local, no API call
Vector search	FAISS, one index per domain
Frontend	Streamlit, custom theme
Hosting	Streamlit Community Cloud, free tier

src/agents.py

The original plan split work across two model tiers — a stronger model for drafting, a lighter one for routing and verification. That fell apart the first time the eval harness actually ran.

20/day
the stronger model's free quota
— burned through mid-run

\$0
total spend — every call now
runs on Flash-Lite

Classify, draft, and verify were consolidated onto Flash-Lite, plus a client-side limiter pacing requests apart. The pipeline itself didn't change — moving to a stronger model on a paid plan is a one-line edit.

A second, independent check before anything ships

ANSWER

The draft is grounded in the source chunks — ships with citations attached.

CLARIFY

The missing detail is about the user's own situation, not something the sources could list — so it asks, instead of guessing.

ESCALATE

No grounded answer is possible — including a correct refusal — so it hands off to a human team.

What this caught during testing

BUG CAUGHT

A correct refusal, marked as a confident answer

The draft accurately said it didn't have enough information — but verification still rated it grounded and high-confidence. Fixed with an explicit rule: a refusal is never an "answer."

BUG CAUGHT

Asked even after already being told

Simplifying the clarify rule to one sentence briefly broke a passing case: the draft asked which role the user held even after the question already said "I'm a manager." Fixed by checking whether the user already stated the detail before asking for it.

TRACK 3 — UNIFIED ENTERPRISE AI AGENT

Why this matters beyond the grade

4

domains covered — HR,
Finance, Support, Legal

48

knowledge-base documents,
authored by hand

20

hand-written eval cases
spanning answer / clarify
/ escalate

0

hallucinated facts observed,
across every run

- ⚠️ Doesn't answer HR, Legal, or Finance questions with false confidence — the main risk of putting an LLM in front of employees and customers.
- 🗑️ Escalates to a human only when it's actually unsure, not on every question.
- 🔑 Runs today on free infrastructure; moving to a stronger, paid model is a config change, not a rewrite.
- 🌐 Deployed and live — a reviewer can try it from a browser, no local setup required.

**Zero cost to build. Zero hallucinations observed.
One line away from a stronger model.**